

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 13****Topics covered :**

1. Linear Dependence and Wronskian
2. General Solutions for homogeneous ODE

- Recall that the Wronskian of two functions y_1 and y_2 which are solutions of the second order ODE

$$\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = 0 \quad (1)$$

is defined as

$$W(y_1(x), y_2(x)) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2 \quad (2)$$

- **Property of Wronskian** [1]: The Wronskian of two functions $y_1(x)$ and $y_2(x)$ that are solutions of (1) in an interval I is either identically zero or never zero in I .

Proof: By definition, the Wronskian of $y_1(x)$ and $y_2(x)$ is given by

$$W(y_1(x), y_2(x)) = y_1 y_2' - y_1' y_2 \quad (3)$$

Taking derivative of (3) gives

$$W' = y_1 y_2'' + y_1' y_2' - y_1'' y_2 - y_1' y_2' = y_1 y_2'' - y_2 y_1'' \quad (4)$$

Since y_1 and y_2 are solutions of (1),

$$y_1'' = -[p(x)y_1' + q(x)y_1] \text{ and } y_2'' = -[p(x)y_2' + q(x)y_2] \quad (5)$$

Substituting (5) in (4) and simplification gives

$$W' = -y_1 [p(x)y_2' + q(x)y_2] + y_2 [p(x)y_1' + q(x)y_1] = -p(x) [y_1 y_2' - y_2 y_1'] = -p(x)W \quad (6)$$

The solution of the Wronskian is thus

$$W = ce^{-\int p(x)dx} \quad (7)$$

Since $e^{-\int p(x)dx} \neq 0$, the Wronskian is 0 if $c = 0$, in which case $W \equiv 0$ over entire I . On other hand, if $c \neq 0$, $W \neq 0$ at any point in the interval I .

- **Linear Independence and Wronskian:** If the wronskian of two solutions y_1 and y_2 is not zero at any point within an interval i.e., if $W(y_1, y_2) \neq 0$ for any $x_0 \in I$, then, y_1 and y_2 are linearly independent on I .

Proof by contradiction: Assume, for the sake of contradiction, that y_1 and y_2 are linearly dependent on the interval I . This implies that there exist constants c_1 and c_2 , not both zero, such that:

$$c_1 y_1(x) + c_2 y_2(x) \equiv 0 \quad (8)$$

Differentiating on both sides of (8) with respect to x gives

$$c_1 y_1'(x) + c_2 y_2'(x) \equiv 0 \quad (9)$$

Now, we can express the system formed by equations (8) and (9) in matrix form as

$$\begin{bmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ for some non-trivial } c_1, c_2 \quad (10)$$

If y_1 and y_2 are linearly dependent this matrix must be singular, meaning its determinant must be zero. However, by the assumption that $W(y_1, y_2) \neq 0$ for any $x_0 \in I$ the Wronskian of y_1 and y_2 is non-zero, which implies that the matrix is of full rank (i.e., non-singular).

This leads to a contradiction, since a non-singular matrix cannot have a non-trivial solution to the homogeneous system. Therefore, our assumption that y_1 and y_2 are linearly dependent must be false.

- **Existence and Uniqueness of Solution to Initial Value Problem:** Let $\{a_n(x), \dots, a_0(x)\}$ and $g(x)$ be continuous on an interval I . Then, for any $x_0 \in I$, the initial value problem $L[y] = g(x)$ and $y^{(i)}(x_0) = y_i$ for $i = 0, 1, \dots, n-1$, has a unique solution on I .
- **Fundamental set of solutions:** The set $\{y_1(x), \dots, y_n(x)\}$ linearly independent solutions of $L[y] = 0$ on the interval I constitute the fundamental set of solutions.
- **Existence of Fundamental Set:** A linear homogeneous ODE $L[y] = 0$ always has a fundamental set of solutions on I .

Proof for $n = 2$: Consider two IVP

$$\begin{aligned} a_2(x) y'' + a_1(x) y' + a_0(x) y &= 0; y(x_0) = 1, y'(x_0) = 0 \\ a_2(x) y'' + a_1(x) y' + a_0(x) y &= 0; y(x_0) = 0, y'(x_0) = 1 \end{aligned} \quad (11)$$

Let $y_1(x)$ and $y_2(x)$ be the unique solutions (due to the uniqueness property) to the IVPs (11). Thus, the Wronskian of y_1 and y_2 at x_0 is given by

$$W(y_1, y_2)(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0 \quad (12)$$

Since, $W(y_1, y_2) \neq 0$, y_1 and y_2 are linearly independent and hence constitute the fundamental set of solutions for $n = 2$.

- **General Solution of homogeneous IVP:** Every Solution $\phi(x)$ for the initial value problem $L[y] = 0$; $y^{(i)}(x_0) = y_i$ can be expressed uniquely as a linear combination of the fundamental set of solutions.

Proof for n=2: Let y_1 and y_2 denote the fundamental set of solution

$$y'' + p(x)y' + q(x)y = 0 \quad (13)$$

and let $\phi(x)$ denote the particular solution to the IVP

$$y'' + p(x)y' + q(x)y = 0; \quad y(x_0) = y_0; \quad y'(x_0) = y_1 \quad (14)$$

Since y_1 and y_2 are fundamental set of solutions, their linear combination, say $y(x)$ is a solution to (13)

$$y(x) = c_1 y_1(x) + c_2 y_2(x) \quad (15)$$

The solution $y(x)$ is a solution to IVP if there exists constants c_1 and c_2 such that

$$\underbrace{\begin{bmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \end{bmatrix} \quad (16)$$

Since y_1 and y_2 are linearly independent ($\implies W(y_1, y_2) \neq 0$ and the matrix $\mathbf{A}$ is full rank), the set of linear equations (16) have a unique solution. Let c_1^* and c_2^* denote the unique solution. Thus,

$$y(x) = c_1^* y_1(x) + c_2^* y_2(x) \quad (17)$$

is a solution to the IVP (14) and so is $\phi(x)$. By uniqueness of the solution, it is necessary that $\phi(x) = y(x) = c_1^* y_1(x) + c_2^* y_2(x)$, which completes the proof.

References

- [1] G. F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd ed. Boca Raton, FL: CRC Press, 2016.